

Clinical Evidence Brief: Once-Weekly Semaglutide in Adults with Overweight or Obesity.

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Background

Obesity is a global health challenge. A glucagon-like peptide-1 (GLP-1) analogue, semaglutide, was approved at doses up to 1 mg once weekly to treat patients with type 2 diabetes and to reduce the risk of cardiovascular events among these patients. Weight loss effects were observed during the phase 2 trial in patients with type 2 diabetes and adults with obesity. Therefore, the global phase 3 Semaglutide Treatment Effect in People with Obesity (STEP) program was conducted to evaluate the efficacy and safety of semaglutide in persons with overweight or obesity at a dose of 2.4 mg weekly with or without weight-related complications, combined with lifestyle intervention. The co-primary end points were the percentage of weight change and weight loss of at least 5% at the end of the trial. The secondary end points were any other improvements observed.

Trial Design

Method overview: 1,961 adults participated in this randomized, double-blind, placebo-controlled clinical trial at 129 sites in 16 countries in Asia, Europe, North America, and South America. Participants were assigned into two groups in a 2:1 ratio and received a dose of 2.4 mg weekly of semaglutide (1,306 participants) or placebo (655 participants), plus a lifestyle intervention for 68 weeks.

Demographics:

1. Gender: 74.1% women.
2. Race: 75.1% white, 13.3% asian, 5.7% black or African American, and 6.0% other.
3. Average age: 46 years old.
4. Average body weight: 105.3 kg.
5. Average body-mass index (BMI, the weight in kilograms divided by the square of the height in meters): 37.9.
6. Average waist circumference: 114.7 cm
7. 43.7% of the participants had pre-diabetes.
8. Most participants (75.0%) had at least one coexisting condition.

Doses: Semaglutide was injected at a dose of 0.25 mg for the first 4 weeks. The dose increased every 4 weeks until reaching 2.4 mg at week 16.

Lifestyle interventions: All participants received individual counseling every 4 weeks on physical activities and diets. Participants recorded the diaries of their daily workout and food they ate.

Results

Co-primary endpoint 1: The semaglutide group achieved a weight loss of -14.9% (-15.3 kg) at week 68, and in the placebo group it was -2.4% (-2.6 kg). The treatment difference was -12.4% (-12.7 kg).

Co-primary endpoint 2: 86.4% of the semaglutide group (1047 participants) lost weight by at least 5% compared to 31.5% of the placebo group (182 participants). Participants in the semaglutide group also had significant improvements in cardiometabolic risk factors and participant-reported physical functioning compared with those in the placebo group.

Confirmatory secondary end points: 69.1% and 12.0% of the semaglutide group and the placebo group, respectively, lost weight more than 10% at week 68. 50.5% and 4.9% of the semaglutide group and the placebo group, respectively, lost weight more than 15% at week 68. Waist circumference reductions were -13.54 cm and -4.13 cm in the semaglutide and placebo groups, respectively. Blood pressure improvement and an increase in physical functioning were also observed.

Supportive secondary endpoints 32.0% and 1.7% of the semaglutide group and the placebo group, respectively, lost more than 20% of their body weight at week 68. Other improvements observed among the participants include glycated hemoglobin levels reverting to normoglycemia and body composition improvements.

Side effects: Similar percentages of participants reported adverse events (89.7% in the semaglutide group, 86.4% in the placebo group). Gastrointestinal disorders (typically nausea, diarrhea, vomiting and constipation) were more frequently reported by the participants in the semaglutide group (74.2%) than the placebo group (47.9%). Most gastrointestinal events were mild-to-moderate in severity and resolved over time. Serious adverse events were reported in 9.8% of semaglutide and 6.4% of placebo participants. More participants discontinued treatment due to gastrointestinal events in the semaglutide group (59, 4.5%) than in the placebo group (5, 0.8%).

Benefits and Risks of the Drug

Benefits: This drug makes it possible for patients with obesity to lose at least 5% of their body weight by week 68 of treatment. The administration of the drug is very convenient, with a dose of 2.4 mg every week.

Risks: Common side effects include nausea and diarrhea, usually transient and mild-to-moderate in severity. Patients should discuss their conditions with their healthcare providers before using the drug. No long-term studies were conducted yet. Therefore, the effects of treatment longer than 68 weeks were unknown.

Strengths and Limitations of the Trial

This randomized, double-blind, and placebo-controlled trial (RCT) was conducted at multiple sites around the world. The results of this single study should be interpreted with real-world evidence (RWE) and a systematic literature review (SLR).

Strengths: This trial used a large sample size and the adherence rate to the trial and the treatment is high (trial completion rate: 94.3% , treatment completion rate: 81.1%, 91.2% of the participants had a body-weight assessment at week 68).

Limitations: The demography in this trial is dominated by women and white participants. The trial duration is relatively short. Persons with type 2 diabetes were excluded in this study. And the participants may have stronger dedication to weight loss than the general population.

Insights for Patient Care

Semaglutide with doses up to 1 mg weekly is approved for the treatment of type 2 diabetes. This trial showed that among adults with overweight or obesity (without diabetes), administering semaglutide at a dose of 2.4 mg weekly with lifestyle intervention has potential to help patients lose 14.9% of their bodyweight, with 86.4% of participants attaining at least 5% weight loss. This study provides a more convenient and effective treatment option compared to the current anti-obesity drugs, which require administration once, twice, or three times every day.

Future Research

While the available evidence is promising, larger studies with longer treatment durations and follow-up periods are needed to provide a more comprehensive understanding of the drug's effects across broader patient populations, including individuals with diabetes, more diverse demographic groups, and adults who are not actively pursuing weight loss.